

Problem

The step responses of a series RLC circuit are

$$v_C = 40 - 10e^{-2000t} - 10e^{-4000t} \text{ V}, \quad t > 0$$

$$i_L(t) = 3e^{-2000t} + 6e^{-4000t} \text{ mA}, \quad t > 0$$

(a) Find C . (b) Determine what type of damping is exhibited by the circuit.

Step-by-step solution

Step 1 of 3

Draw the series RLC circuit.

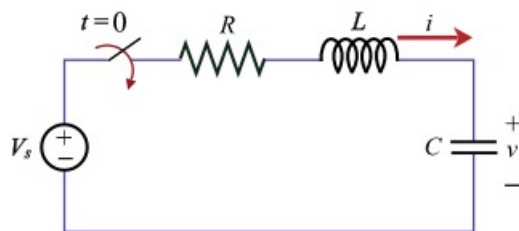


Figure 1

Step 2 of 3

Consider the following step responses of the series RLC circuit:

The voltage across the capacitor is $v_C = 40 - 10e^{-2000t} - 10e^{-4000t}$ V, $t > 0$

The current through the inductor is $i_L(t) = 3e^{-2000t} + 6e^{-4000t}$ mA, $t > 0$

(a)

As a series RLC circuit same current flowing through the each element. Therefore the capacitor current is equal to the inductor current.

$$\begin{aligned} i_C(t) &= i_L(t) \\ &= 3e^{-2000t} + 6e^{-4000t} \text{ mA, } t > 0 \end{aligned}$$

Calculate the value of the capacitor using the following formula.

$$\begin{aligned} i_C(t) &= C \frac{dv_C}{dt} \\ C &= i_C(t) \left(\frac{1}{\left(\frac{dv_C}{dt} \right)} \right) \end{aligned}$$

Calculate the derivative of the capacitor voltage.

$$\begin{aligned} \frac{d}{dt}(v_C) &= \frac{d}{dt}(40 - 10e^{-2000t} - 10e^{-4000t}) \\ &= ((-10(-2000)e^{-2000t}) - 10(-4000e^{-4000t})) \\ &= 20000e^{-2000t} + 40000e^{-4000t} \\ &= (2 \times 10^4)(e^{-2000t} + 2e^{-4000t}) \end{aligned}$$

Substitute $3e^{-2000t} + 6e^{-4000t}$ mA for $i_C(t)$ and $(2 \times 10^4)(e^{-2000t} + 2e^{-4000t})$ for $\frac{dv_C}{dt}$ to calculate C.

$$\begin{aligned} C &= ((3e^{-2000t} + 6e^{-4000t}) \times 10^{-3}) \left(\frac{1}{(2 \times 10^4)(e^{-2000t} + 2e^{-4000t})} \right) \\ &= (3 \times 10^{-3}(e^{-2000t} + 2e^{-4000t})) \left(\frac{1}{(2 \times 10^4)(e^{-2000t} + 2e^{-4000t})} \right) \\ &= \frac{(3 \times 10^{-3})(e^{-2000t} + 2e^{-4000t})}{(2 \times 10^4)(e^{-2000t} + 2e^{-4000t})} \\ &= 1.5 \text{ nF} \end{aligned}$$

Thus, the value of the capacitance is 1.5 nF.

Step 3 of 3

(b)

Form the step responses of the series RLC circuit poles of the circuit are, $s_1 = -2000$ and $s_2 = -4000$. These are real and negative; hence the circuit exhibited an **Overdamped** case.